

Predictive State Representations for Continual Reinforcement Learning: An Exploratory Study

Observable GVs as a compact predictive state under partial observability and hidden non-stationarity

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Introduction and Methods

Motivation

From Observation History to Predictive State

Online continual learning agents must learn and act from an ongoing stream of experience, without restarting or relying on a fixed environment. In this setting, state representations need to be **both control-relevant and continually adaptable**. GVF-based predictive states offer a way to represent what the agent expects to happen next while those predictions are learned online. **We study whether these predictive states improve control, support adaptation under change, and which predictions matter most.**

Core Hypothesis

State should represent what the past predicts about the future, not merely store the past.

A useful state should summarize not only **what the agent has recently observed**, but also **what is likely to happen next**. However, not every prediction is equally useful.

Experiments and Conclusions

RQ1 | Combined predictive state tops all tasks

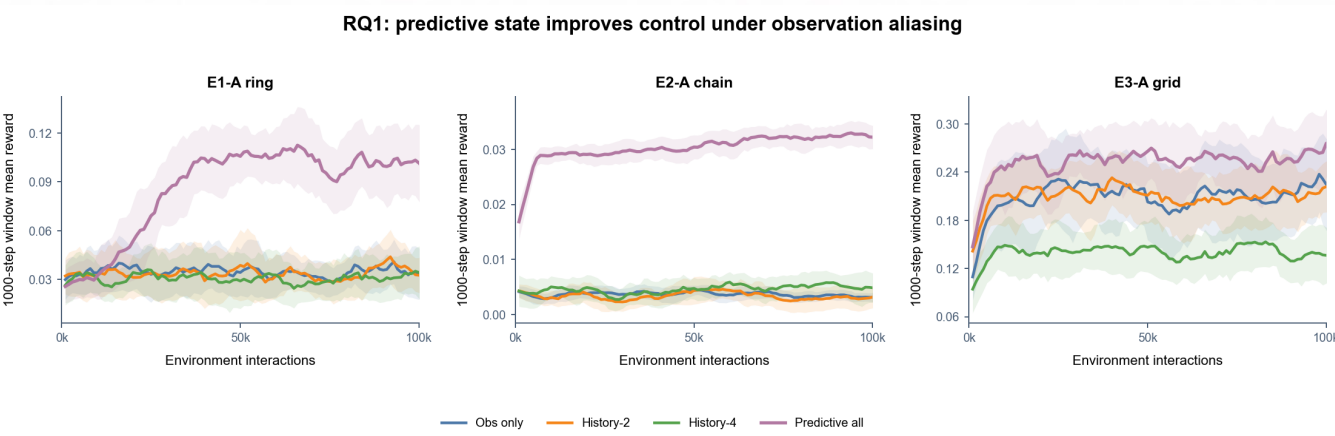


Figure 1. Windowed reward (mean and approximate 95% CI) separates most strongly in the ring and chain.

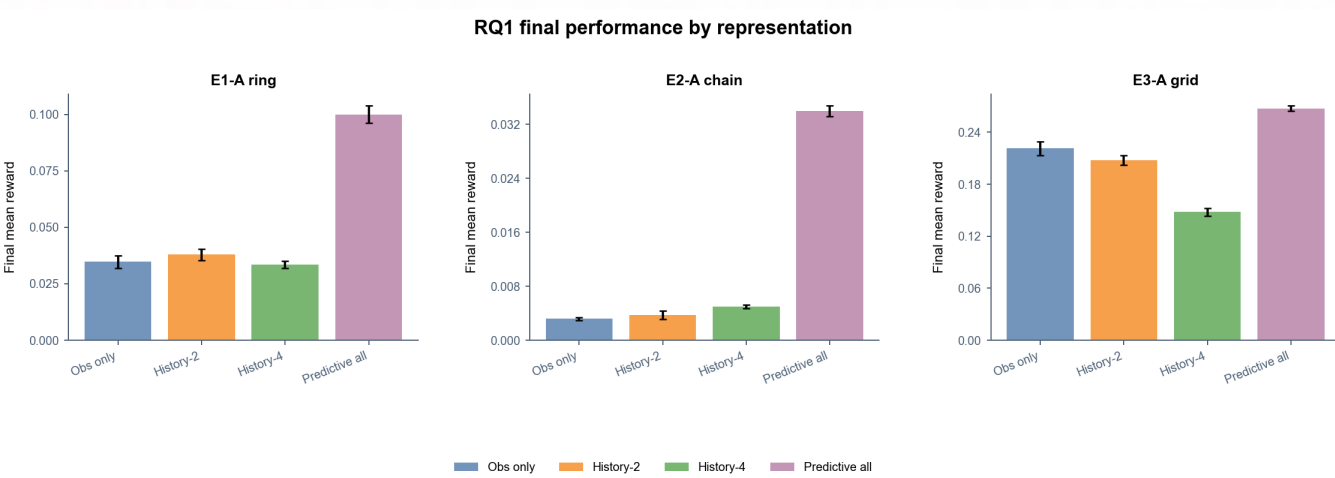


Figure 2. Predictive all achieves the highest final mean reward in E1-A, E2-A, and E3-A.

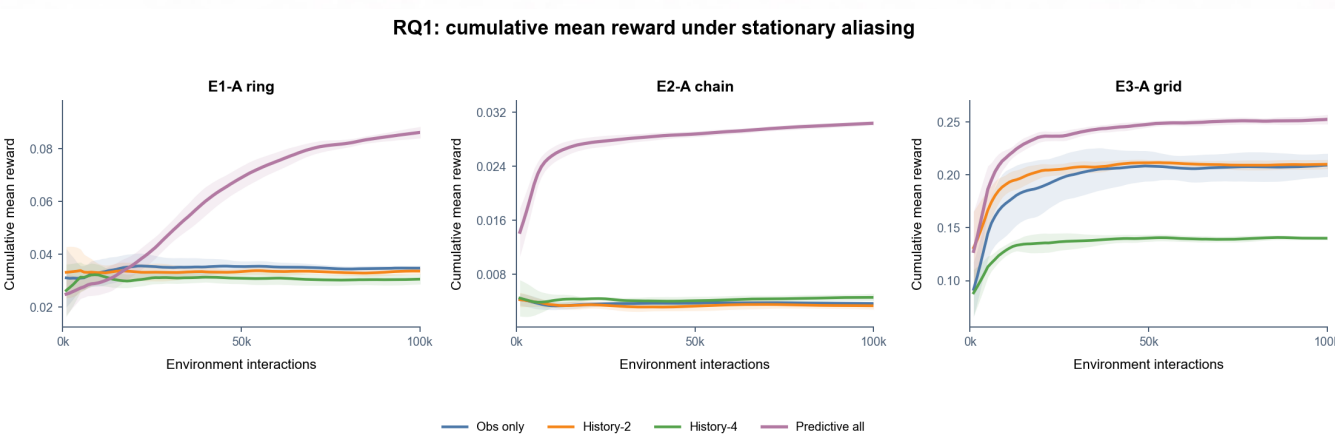


Figure 3. Cumulative reward confirms a persistent advantage, rather than a late-window artifact.

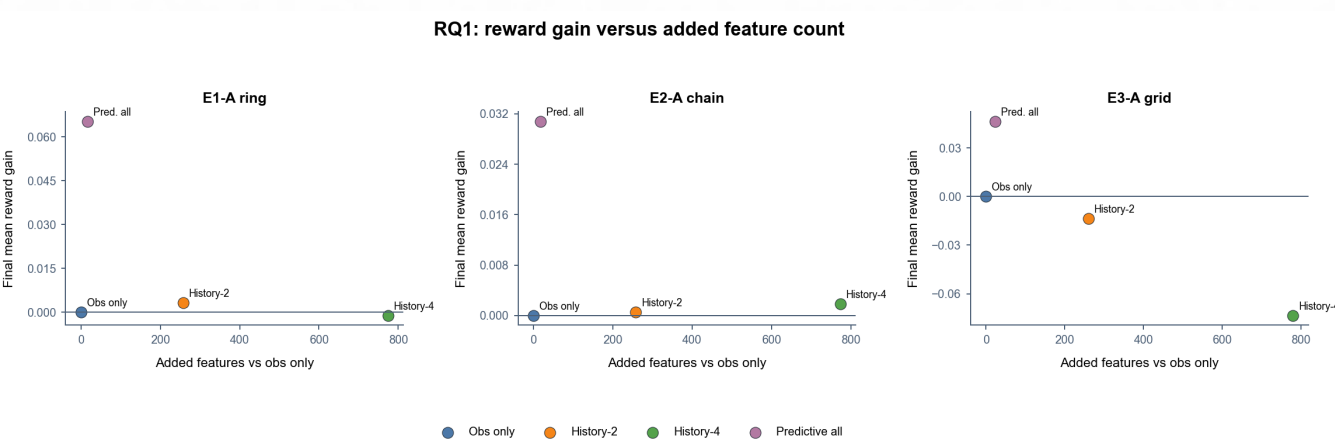
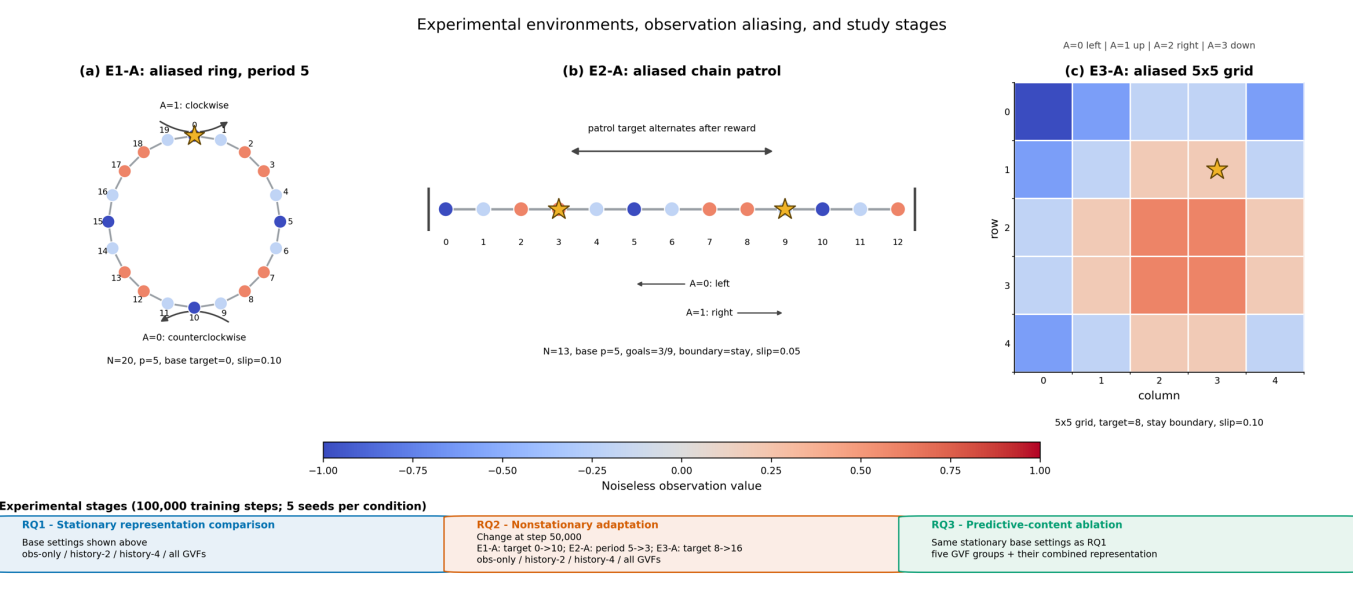


Figure 4. Fixed history adds hundreds of features without matching the predictive state's reward gain.

Conclusion

- Predictive state representations provide a more effective and compact solution than finite history stacking.
- Predictive state representations achieves the highest final mean reward, while using less features.

Environments



Env.	Topology	Actions	Goal	Step cost	Structure
E1-A	20-state ring	2	Fixed: 0	-0.01	cyclic symmetry
E2-A	15-state chain	2	Patrol: 3 / 11	-0.001	long-range dependence
E3-A	5x5 grid	4	Fixed: (1,3)	-0.01	two-dimensional structure

Shared setting: aliased noisy observations ($\sigma = 0.10$), reward +1 at the goal, and continuing tasks without reset.

RQ2 | Predictive state leads after hidden change

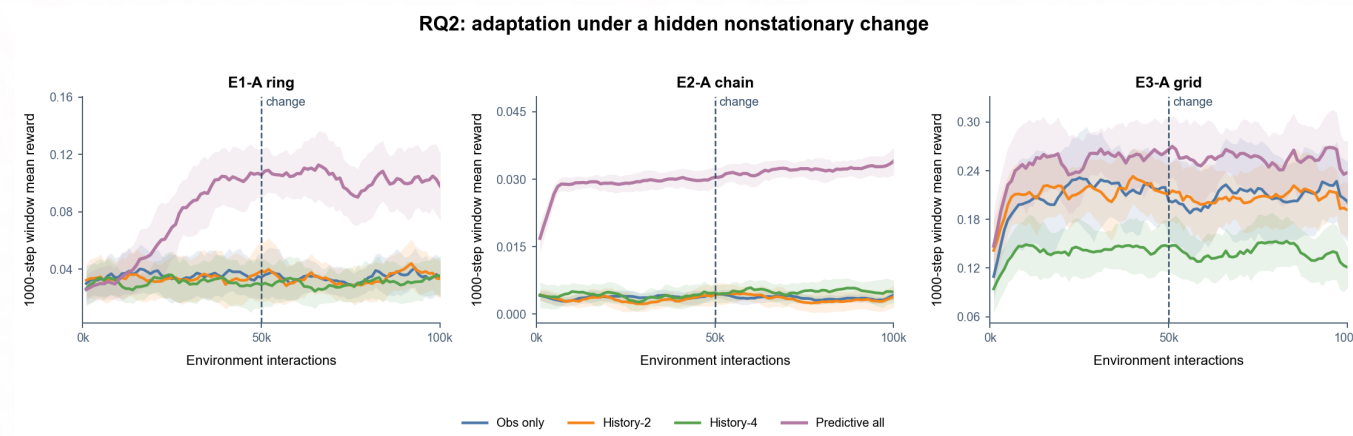


Figure 5. A hidden parameter changes at 50k; predictive all remains highest before and after the change.

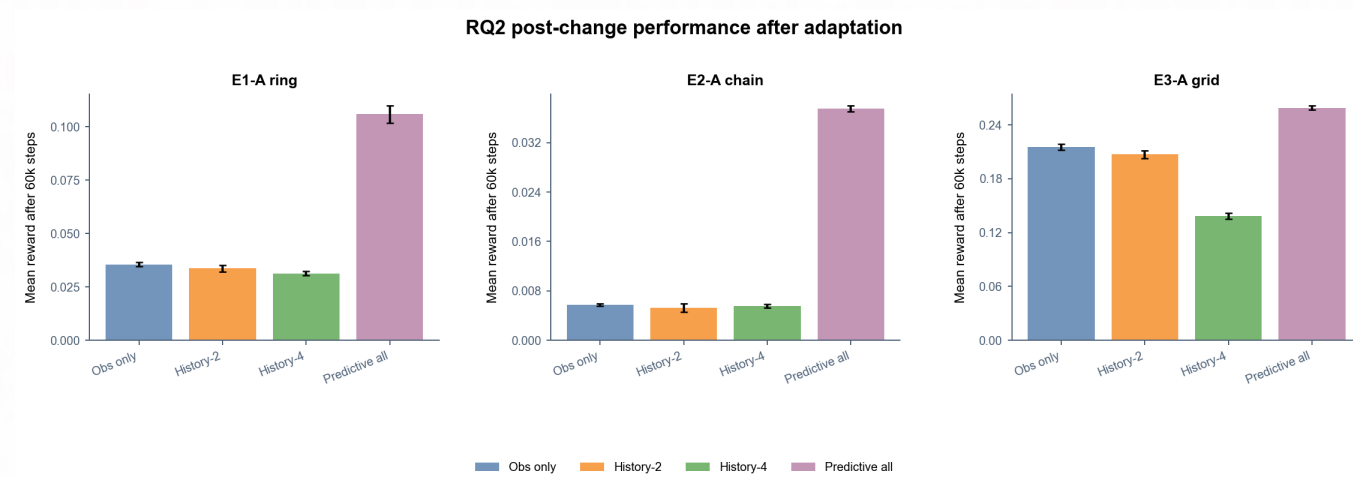


Figure 6. Post-change performance preserves the same ordering across all three scenarios.

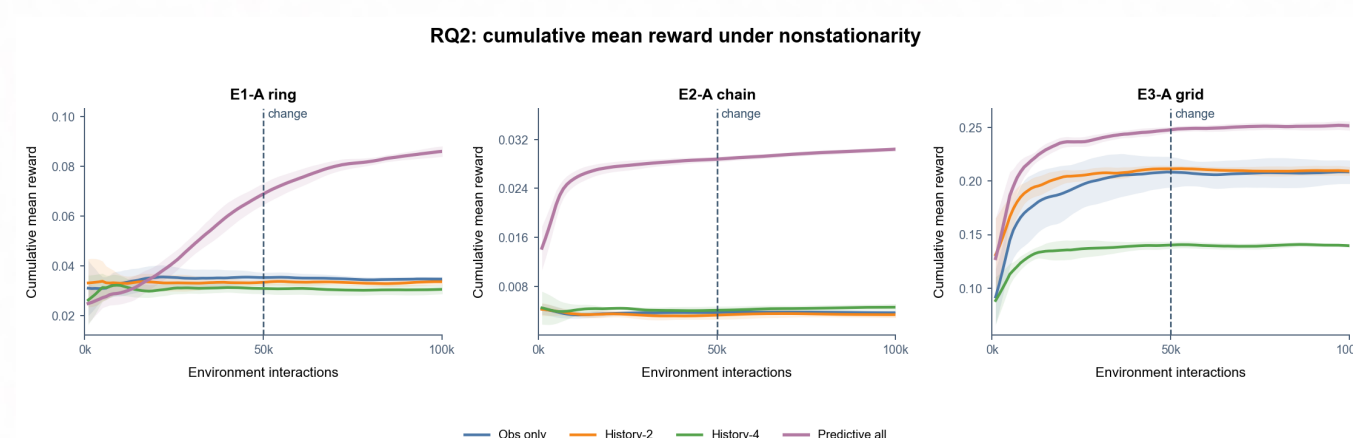


Figure 7. The cumulative advantage persists through the nonstationary phase.

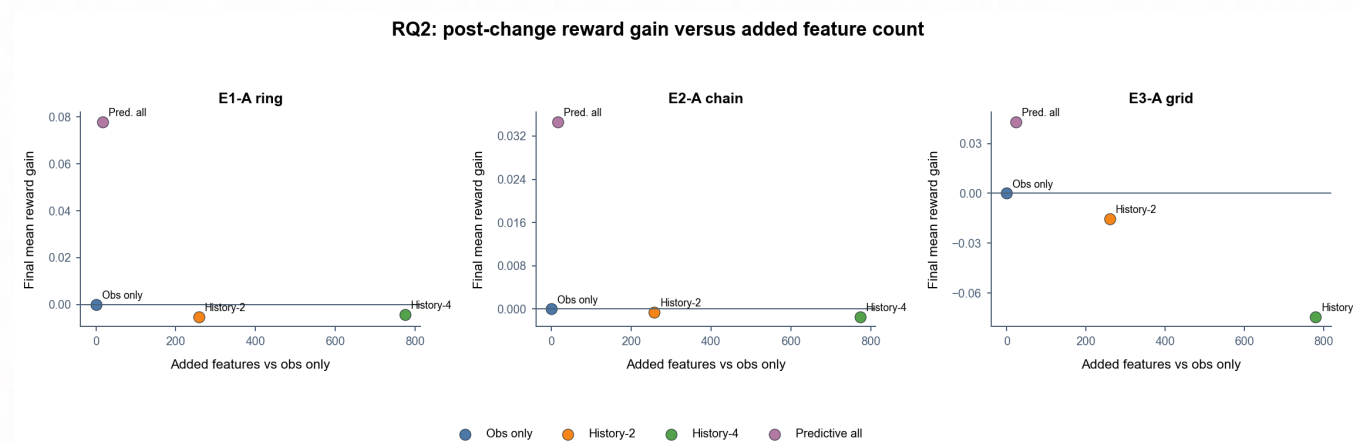
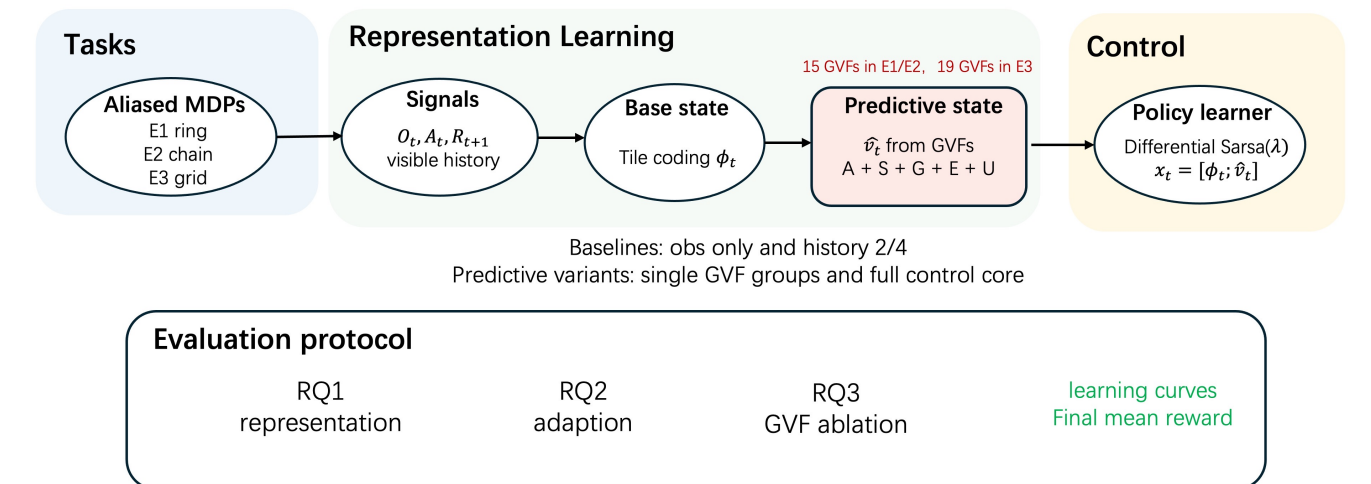


Figure 8. After the hidden change, fixed history adds hundreds of features without matching the predictive state's reward gain.

Conclusion

- Predictive representations are more robust to nonstationary changes.
- Predictive states maintain stronger control performance under hidden non-stationarity.

Methods



GVFs

Symbol	Predictive content	GVF questions
$\hat{V}_t^A$	Action consequence	Short- and long-horizon reward consequences for each action
$\hat{V}_t^S$	Successor occupancy	Future observation and observation magnitude at two horizons
$\hat{V}_t^G$	Goal progress	Short- and long-horizon goal-reward predictions
$\hat{V}_t^E$	Event termination	Event occurrence and expected time to event
$\hat{V}_t^U$	Observation change	Short-/long-horizon observation change and reward-conditioned change

RQ3 | Useful predictive content is task-dependent

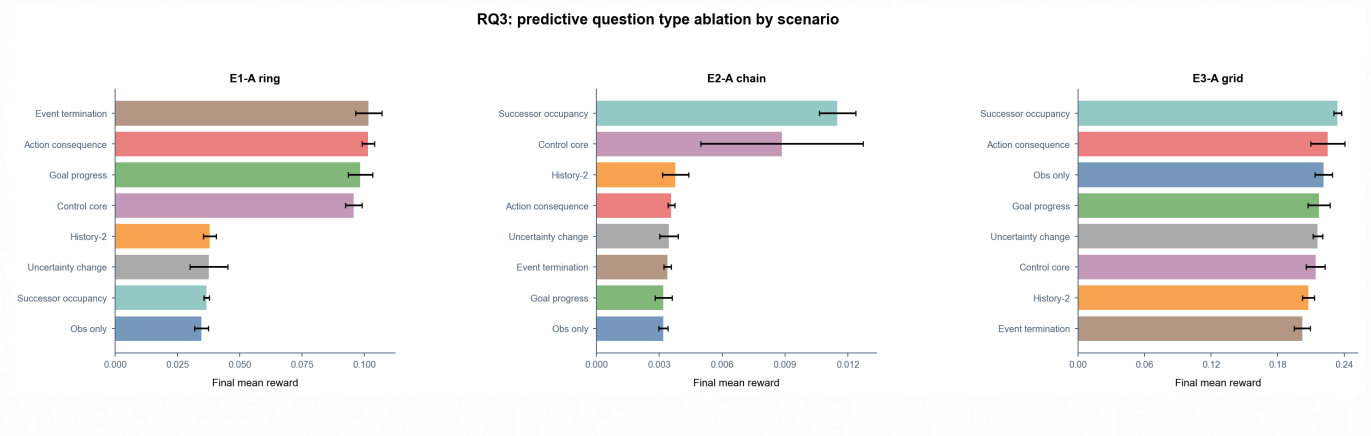


Figure 9. The best single predictive question changes with the task structure.

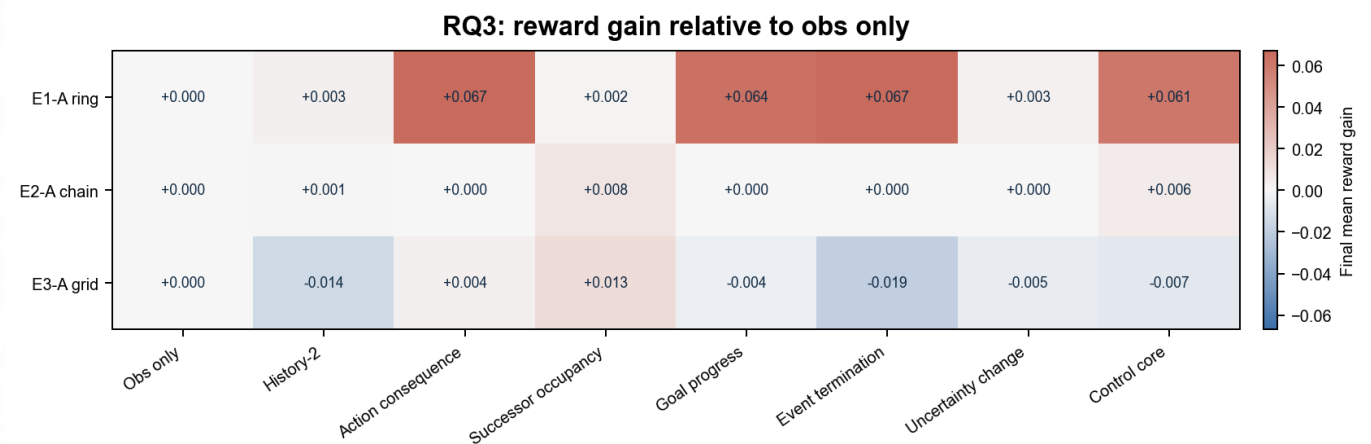


Figure 10. Gains over obs only are broad in E1, concentrated in E2, and selective in E3.

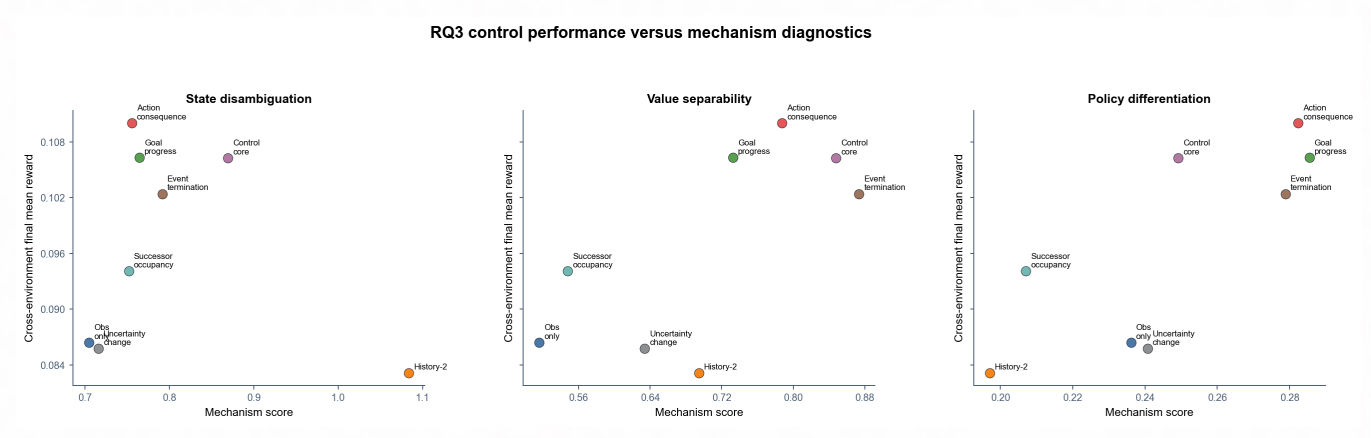


Figure 11. Relationship between control performance and three mechanism diagnostics.

Conclusion

- Not all predictions help.
- Combining predictions is most robust.
- Different environment matters. Action / Event predictions stand out in E1-A, while future consequence is the best in E2-A / E3-A.

Next Step

Next step will move beyond manually specified predictive questions by automatically learning or selecting GVFs as the environment changes. These GVFs should also remain interpretable, with each prediction linked to meaningful events, future consequences, and control decisions. This would make predictive state representations both adaptive and easier to understand, via cumulative ablation analysis.

Across three aliased environments and three control studies, GVF-based state representation generally outperformed observation and fixed history. However, GVF ablations showed that useful predictive content is task-dependent: event and action predictions benefited the ring, while successor occupancy was strongest in the chain and grid.